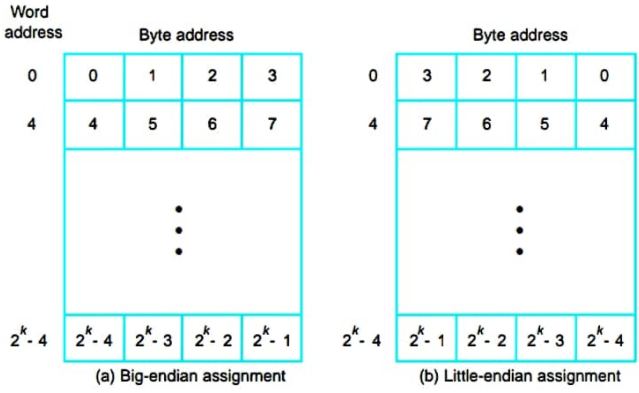
Some Basic Concepts

Basic Concepts

- The maximum size of the memory that can be used in any computer is determined by the addressing scheme.
  16-bit addresses = 2^{16} = 64K memory locations
- Most modern computers are byte addressable.



Traditional Architecture

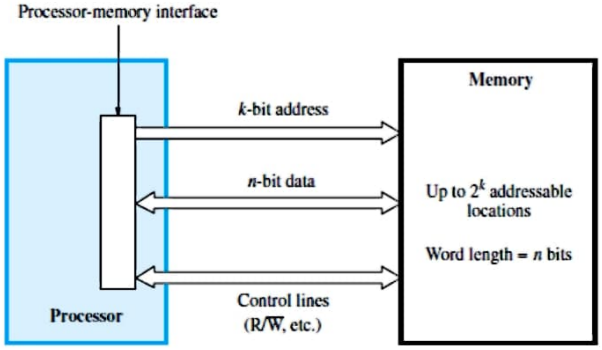
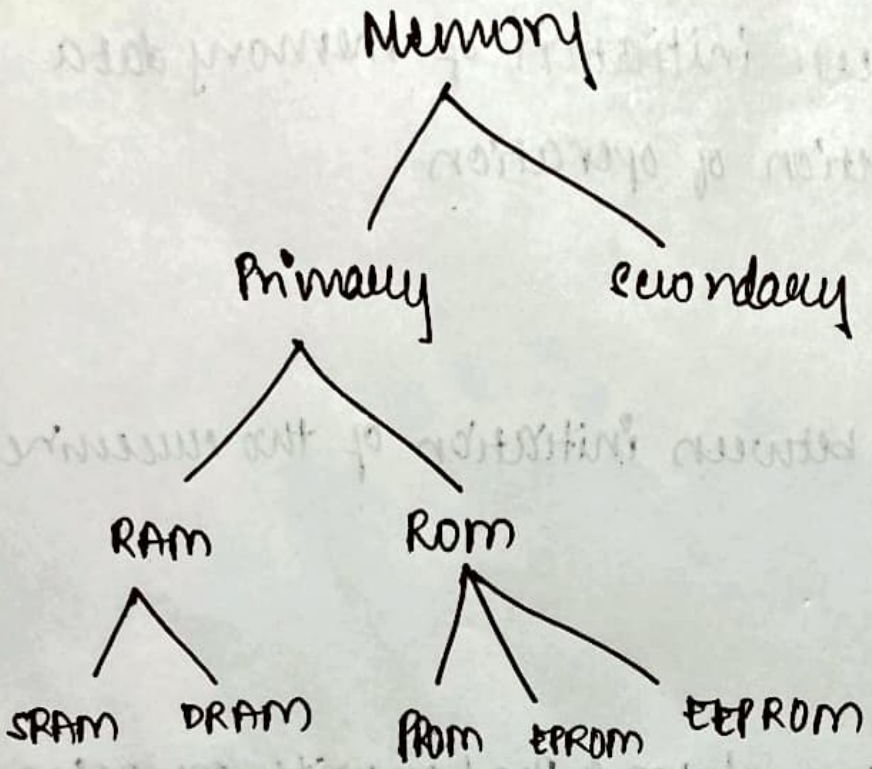
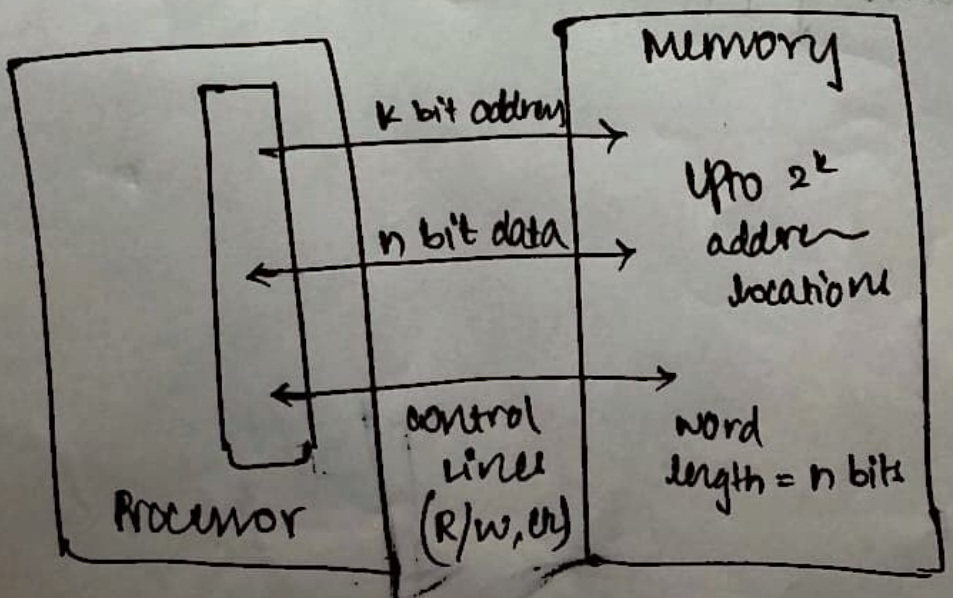


Figure 8.1 Connection of the memory to the processor.



Maximum size of memory that can be used is determined by addressing scheme.

16 Bit address = 2^{16} = 64 K memory locations



Memory acces time

time interval between initiation of memory data transfer and completion of operation

Memory cycle time

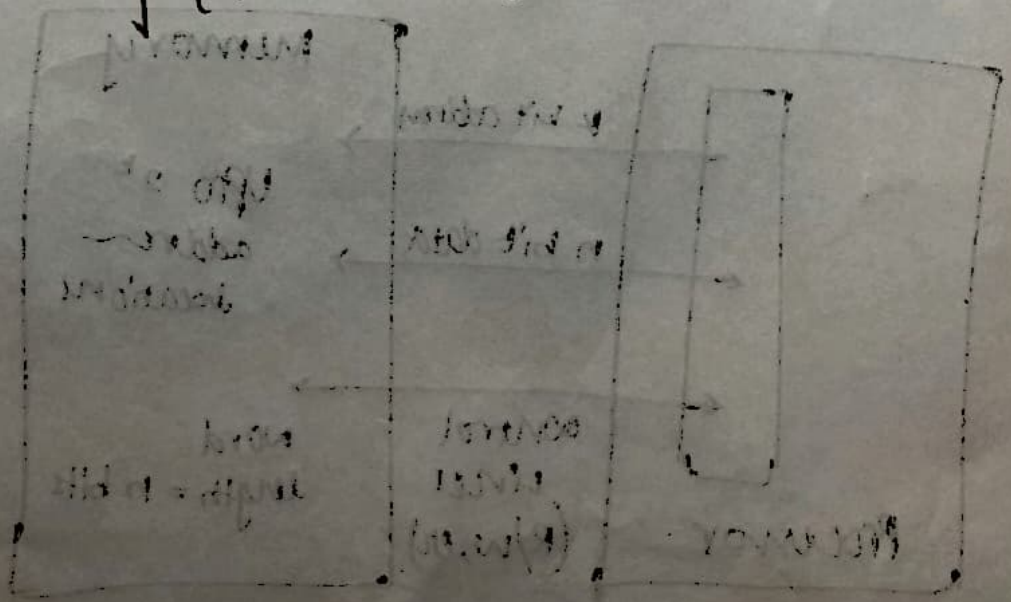
minimum time delay between initiation of two successive memory operations.

RAM

Any location can be accessed for a Read or write operation in some fixed amount of time that is independent of the locations address.

cache memory (increase effective speed)

virtual memory (increase effective size)



Some basic concepts(Contd.,)

- Measures for the speed of a memory:
  - memory access time: time interval between the initiation of memory data transfer and the completion of the operation
  - memory cycle time: minimum time delay required between the initiation of two successive memory operations. Eg: time between two successive Read operarions
- An important design issue is to provide a computer system with as large and fast a memory as possible, [illegible] cost target.
- Several techniques to increase the effective size and speed of the memory:
  - Cache memory (to increase the effective speed).
  - Virtual memory (to increase the effective size).
- "Block transfer" – bulk data transfer - a critical parameter for the performance of the main memory is its ability to read or write blocks of data at high speed.

Semiconductor RAM Memories

Internal organization of memory chips

- RAM – any location can be accessed for a Read or Write operation in some fixed amount of time that is independent of the location's address.
- Each memory cell can hold one bit of information.
- Memory cells are organized in the form of an array.
- One row is one memory word.
- All cells of a row are connected to a common line, known as the "word line".
- Word line is connected to the address decoder.
- Sense/write circuits are connected to the data input/output lines of the memory chip.

16 X 8 organization

16 words of 8 bit each

4 Bits address (A_0 A_1 A_2 A_3)

8 Data lines

2 Vcc / GND

2 Chipselect and R/W

= 16 external connections

128 X 8

7 Bits address $[2^7 = 128]$

8 Data lines

2 Vcc / GND

2 Chipselect and R/W

1K X 1

$[2^{10} = 2^5 \times 2^5]$

10

1

2

2

15

1024 cells

1K bits

1024 bits

128 x 8

$2^7 \times 8$

128 rows x 8 col

7 bit address

8 data line

2 Vcc / GND

2 CS & R/W

19 external connection

1K x 1

1K

$2^{10} \times 1 = 2^5 \times 2^5 = 32 \times 32$

Rows = 2^5 = 32 ⇒ 5 bits for row

Col = $2^5/1$ = 32 ⇒ 5 bits for col

10 address line

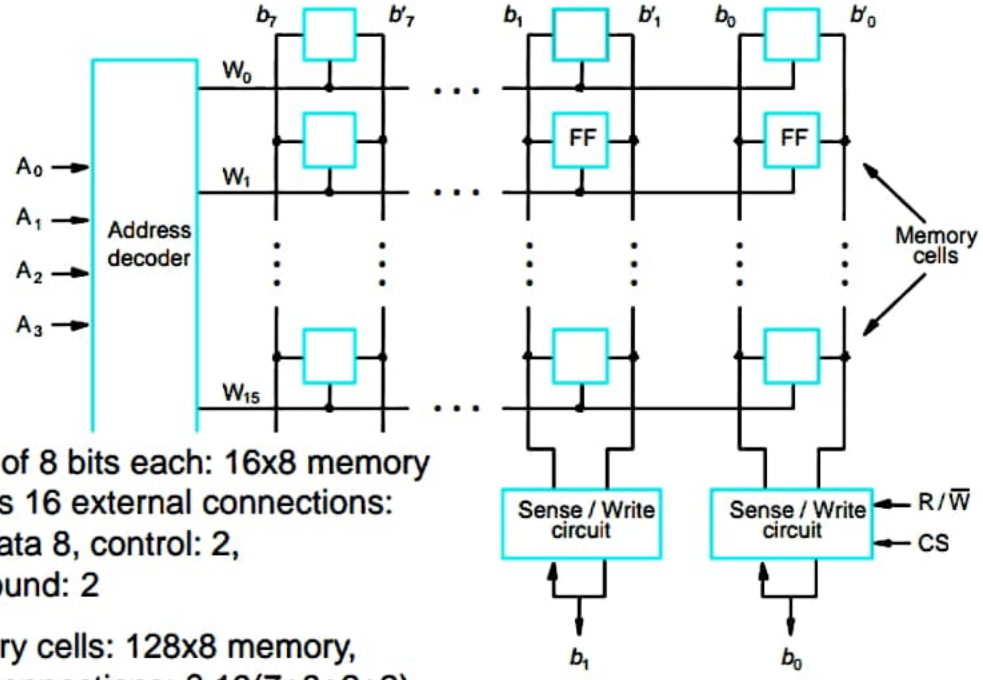
1 data line

2

2

15 ext connection

Internal Organization of Memory Chips



16 words of 8 bits each: 16x8 memory org.. It has 16 external connections: addr. 4, data 8, control: 2, power/ground: 2

1K memory cells: 128x8 memory, external connections: ? 19(7+8+2+2)

1Kx1:? 15 (10+1+2+2)

Figure 8.2. Organization of bit cells in a memory chip.

A Memory Chip

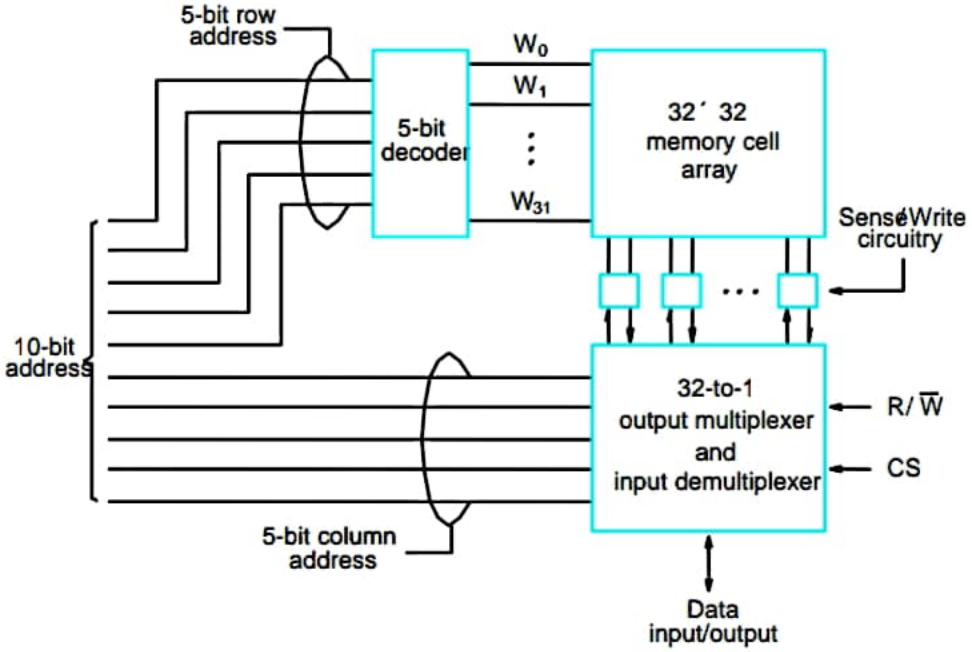


Figure 8.3. Organization of a 1K × 1 memory chip.

8.2.2 Static Memories

Memories that consist of circuits capable of retaining their state as long as power is applied are known as *static memories*. Figure 8.4 illustrates how a *static RAM* (SRAM) cell may be implemented. Two inverters are cross-connected to form a latch. The latch is connected to two bit lines by transistors T_1 and T_2. These transistors act as switches that can be opened or

closed under control of the word line. When the word line is at ground level, the transistors are turned off and the latch retains its state. For example, if the logic value at point X is 1 and at point Y is 0, this state is maintained as long as the signal on the word line is at ground level. Assume that this state represents the value 1.

Static Memories

- The circuits are capable of retaining their state as long as power is applied.

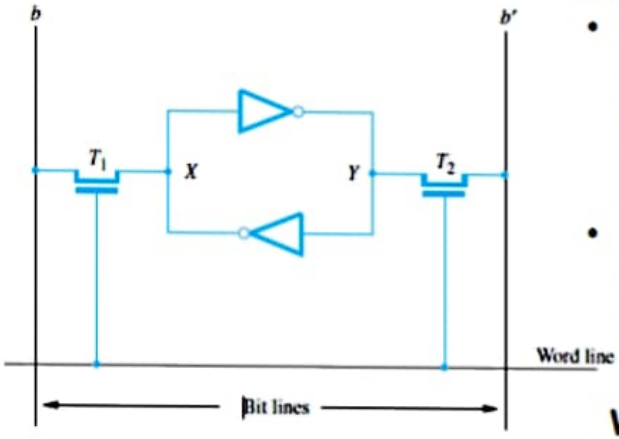


Figure 8.4 A static RAM cell.

Read Operation

- In order to read the state of the SRAM cell, the word line is activated to close switches *T*1 and *T*2. If the cell is in state 1, the signal on bit line *b* is high and the signal on bit line *b*' is low.
- The Sense/Write circuit at the end of the two bit lines monitors their state and sets the corresponding output accordingly.

Write Operation

- During a Write operation, the Sense/Write circuit drives bit lines *b* and *b*', instead of sensing their state.
- It places the appropriate value on bit line *b* and its complement on *b* and activates the word line.

Static Memories

- CMOS cell: low power consumption

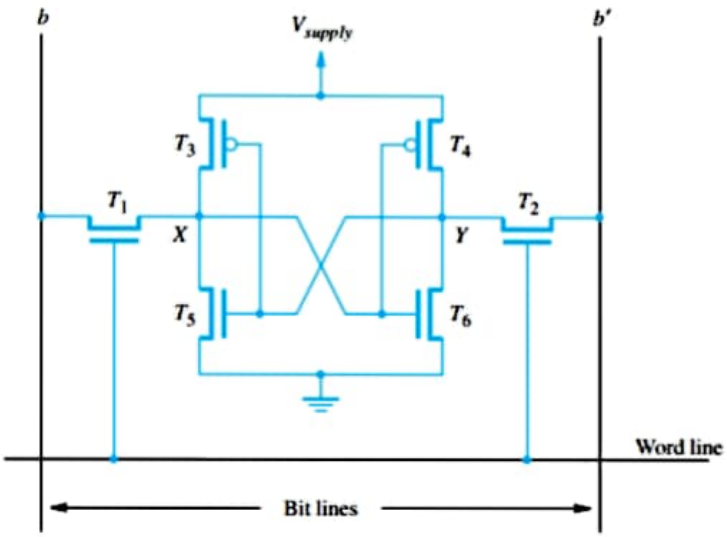


Figure 8.5 An example of a CMOS memory cell.

CMOS Cell

A CMOS realization of the cell in Figure 8.4 is given in Figure 8.5. Transistor pairs (T_3, T_5) and (T_4, T_6) form the inverters in the latch (see Appendix A). The state of the cell is read or written as just explained. For example, in state 1, the voltage at point X is maintained high by having transistors T_3 and T_6 on, while T_4 and T_5 are off. If T_1 and T_2 are turned on, bit lines b and b' will have high and low signals, respectively.

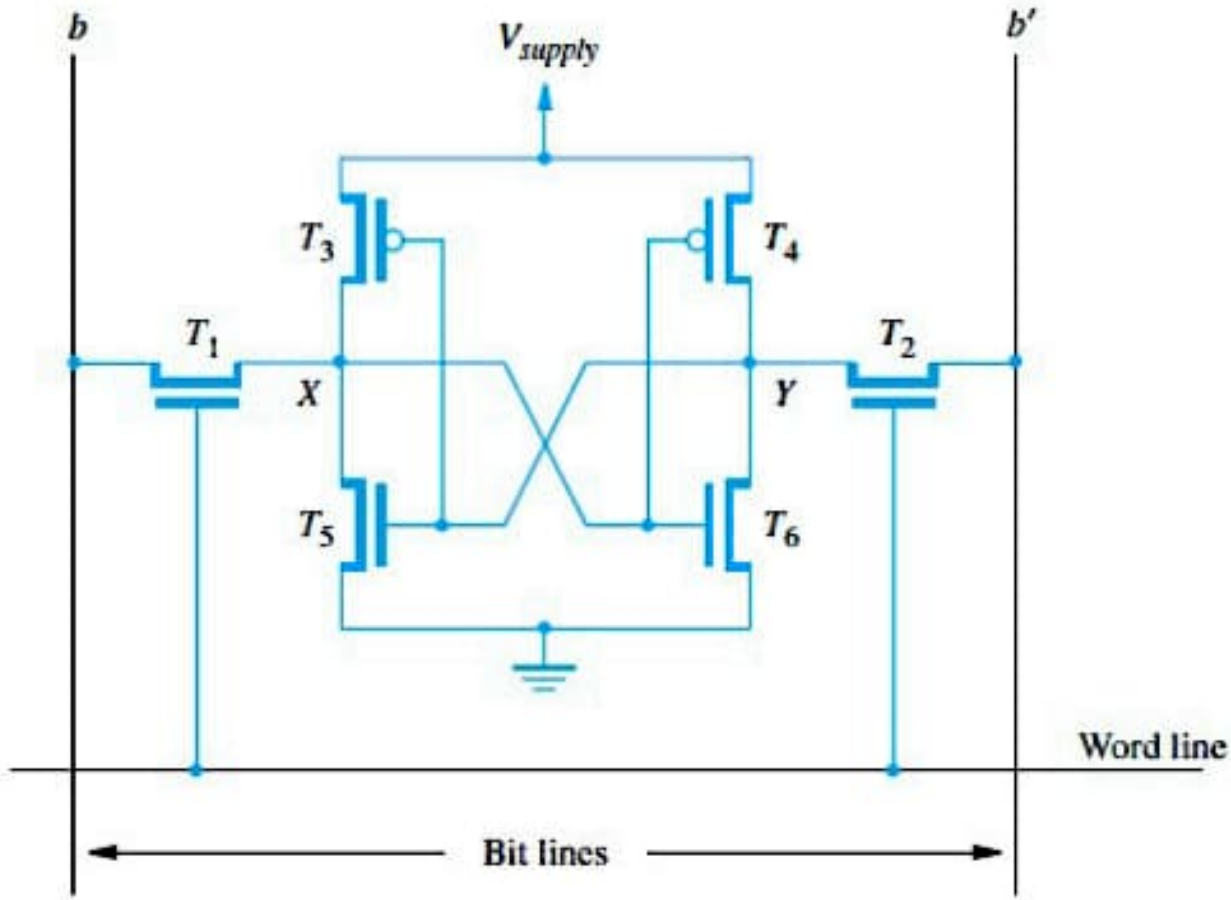


Figure 8.5 An example of a CMOS memory cell.

8.2.3 Dynamic RAMs

Static RAMs are fast, but their cells require several transistors. Less expensive and higher density RAMs can be implemented with simpler cells. But, these simpler cells do not retain their state for a long period, unless they are accessed frequently for Read or Write operations. Memories that use such cells are called *dynamic RAMs* (DRAMs).

Information is stored in a dynamic memory cell in the form of a charge on a capacitor, but this charge can be maintained for only tens of milliseconds. Since the cell is required to store information for a much longer time, its contents must be periodically *refreshed* by restoring the capacitor charge to its full value. This occurs when the contents of the cell are read or when new information is written into it.

An example of a dynamic memory cell that consists of a capacitor, C, and a transistor, T, is shown in Figure 8.6. To store information in this cell, transistor T is turned on and an appropriate voltage is applied to the bit line. This causes a known amount of charge to be stored in the capacitor.

After the transistor is turned off, the charge remains stored in the capacitor, but not for long. The capacitor begins to discharge. This is because the transistor continues to

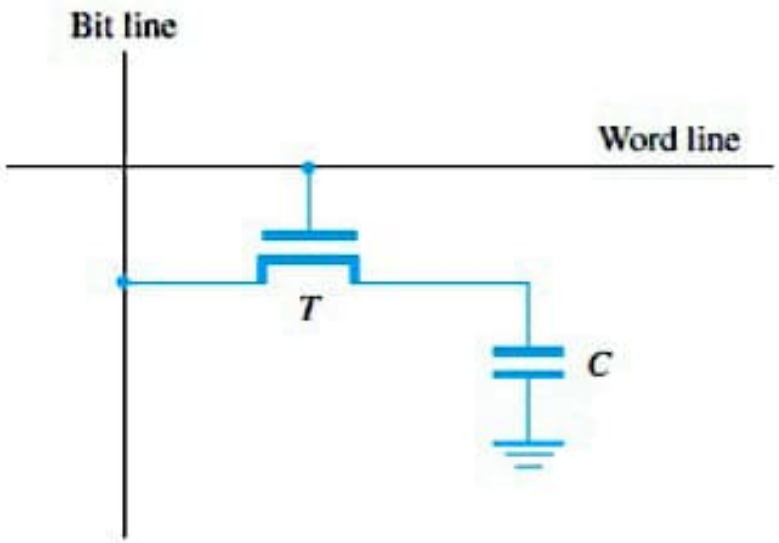


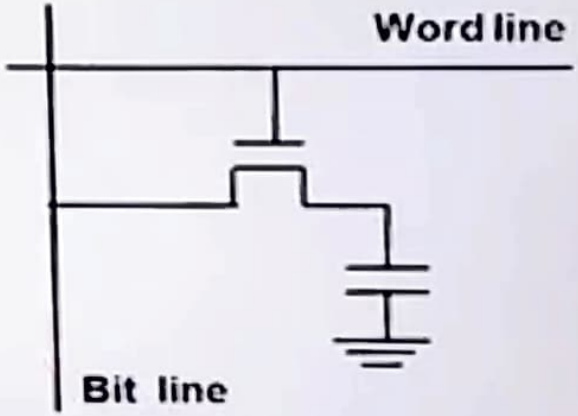
Figure 8.6 A single-transistor dynamic memory cell.

conduct a tiny amount of current, measured in picoamperes, after it is turned off. Hence, the information stored in the cell can be retrieved correctly only if it is read before the charge in the capacitor drops below some threshold value. During a Read operation, the transistor in a selected cell is turned on. A sense amplifier connected to the bit line detects whether the charge stored in the capacitor is above or below the threshold value. If the charge is above the threshold, the sense amplifier drives the bit line to the full voltage representing the logic value 1. As a result, the capacitor is recharged to the full charge corresponding to the logic value 1. If the sense amplifier detects that the charge in the capacitor is below the threshold value, it pulls the bit line to ground level to discharge the capacitor fully. Thus, reading the contents of a cell automatically refreshes its contents. Since the word line is common to all cells in a row, all cells in a selected row are read and refreshed at the same time.

ASYNCHRONOUS DRAMS

- The timing of the memory device is controlled asynchronously.
- The device connected to this memory is responsible for the delay.
- Address lines are divided into two parts and multiplexed.
  - Upper half of address:
    - Loaded into Row Address Latch using Row Address Strobe(RAS).
  - Lower half of address:
    - Loaded into Column Address Latch using Column Address Strobe(CAS).

1-Transistor Memory Cell (DRAM)



- Write Operation:
  - 1. Activate the word line.
  - 2. Transistor in selected cell is turned ON
  - 3.Capacitor gets charged to the voltage supplied to bit line.
- Read Operation:
  - 1. Activate the word line.
  - 2.Transistor in selected cell is turned ON
  - 3. Cell and bit line share charges
  - Capacitor > Threshold voltage, Bit line=1
  - Capacitor < Threshold voltage, Bit line= 0
- Refresh
  - 1. Just do a dummy read to every cell.

32M X 8

= 256M Bits

can be written as

= 16K X 16K

= 16K

$= 2^4 \times 2^{10}$

= 14 address bits for row

For Column

= 16K

$= 16 \times 2^{10}$

$= 16 \times 1024$

$= \frac{16 \times 1024}{8}$

(Because we made group of 8 bits)

= 2048

$= 2^{11}$

= 11 bits for column

∴ Rows = 2^{14}

Col = $2^{14}/8 = 2^{11}$ (bytes)

Each row can store 2048 bytes of data.

= 2048 bytes

∴ No of row bits = 14

No of col bits = 11

∴ Tot address bits = 25

$A_{24} - A_{11} / A_{10} - A_0$

A Dynamic Memory Chip

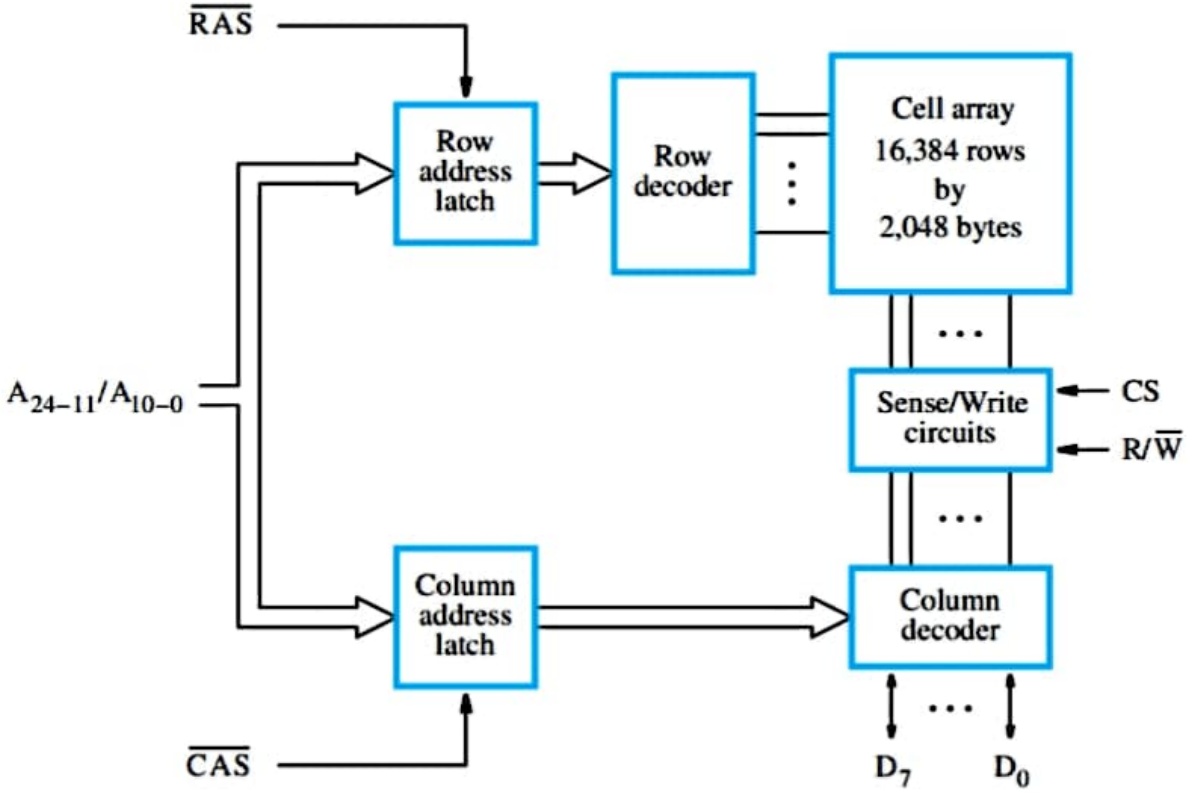


Figure 8.7 Internal organization of a 32M × 8 dynamic memory chip.

Design 16 M bit DRAM chip(2M X 8 memory organization)

$16\ 2^{20}$

$2^4\ 2^{20}$

2^{24}

$2^{12} \times 2^{12}$ Equivalent to 4K x 4K

ROWS= 2^{12} =4096

COLUMNS=4096/8=512 columns (8 bits)

No: of bits required for ROW Add. $X=\log_2 2^{12}=12$

No: of bits required for COLUMN Add. $Y=\log_2 2^9=9$

Total address bit n=x(higher order) + y(lower order)

=12 + 9 =21 bits

$=A_{20}\text{-}A_9 + A_8\text{-}A_0$

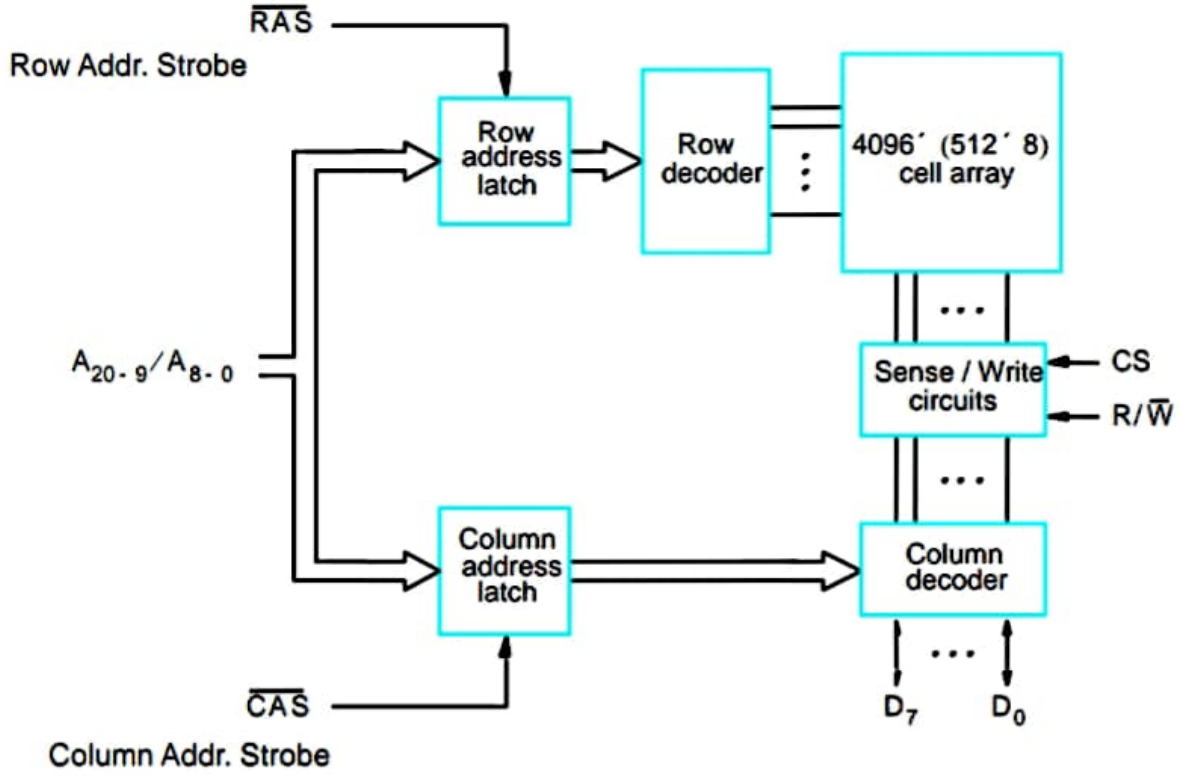


Figure 5.7. Internal organization of a 2M X 8 dynamic memory chip.

FAST PAGE MODE

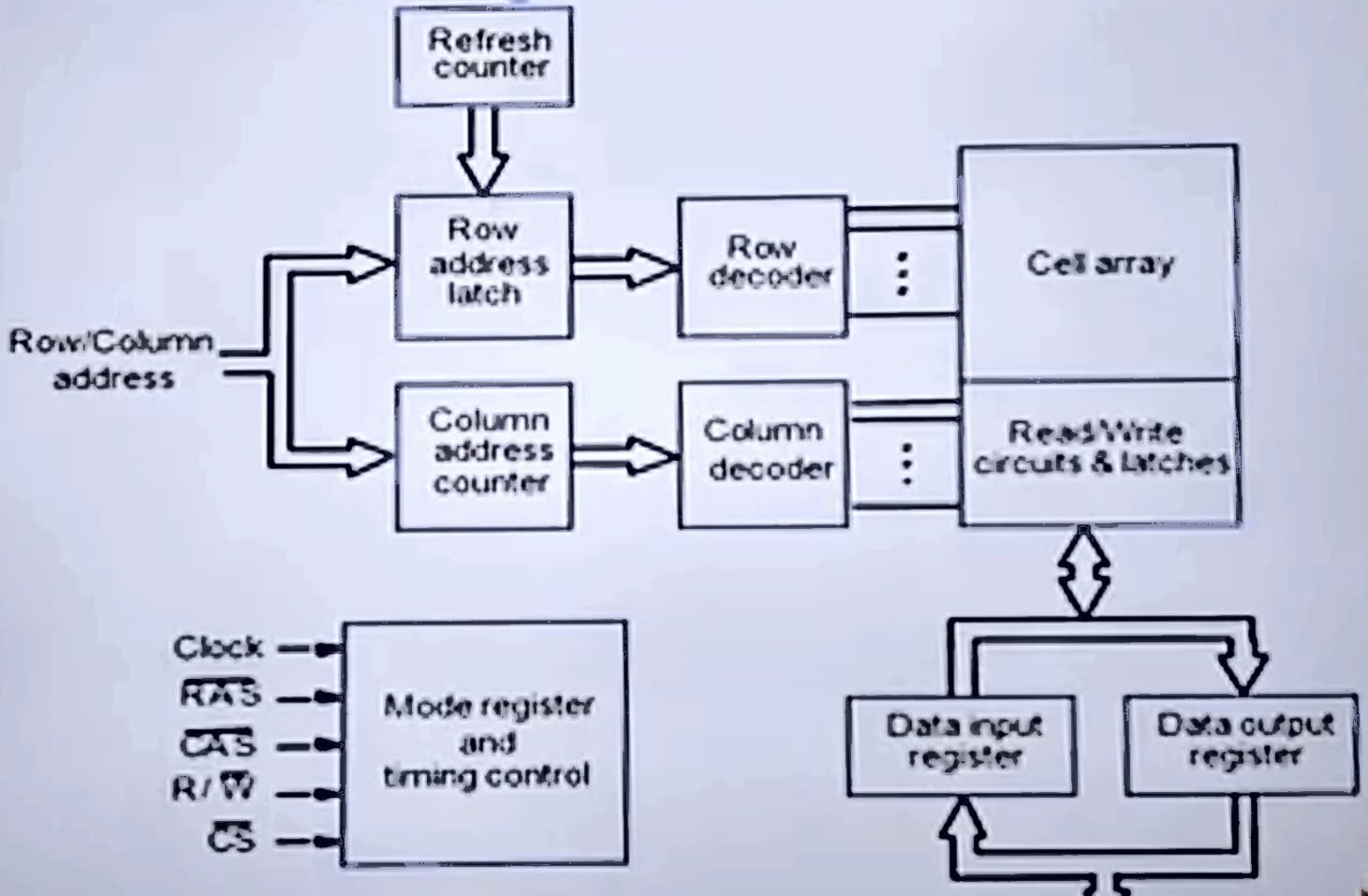
- When the DRAM in last slide is accessed, the contents of all 4096 cells in the selected row are sensed, but only 8 bits are placed on the data lines D7-0, as selected by A8-A0.
- Suppose if we want to access the consecutive bytes in the selected row.
- This can be done without having to reselect the row.
  - Add a latch at the output of the sense circuits in each row.
  - All the latches are loaded when the row is selected.
  - Consecutive bytes can be transferred by applying consecutive column addresses under the control of successive CAS signals..
  - This allows a block of data to be transferred at a much faster rate than random accesses.
  - A small collection/group of bytes is usually referred to as a **block.**
- This transfer capability is referred to as the **FAST PAGE MODE ACCESS**.
- Good for bulk transfer



SYNCHRONOUS DRAMS(SDRAM)

- SDRAM is the commonly used name for various kinds of dynamic RAM that are synchronized with processor clock.
- The structure of this memory is the same as in Asynchronous DRAMs.
- Instead of a column address Latch, there is a column address Counter.
- The outputs of the sense circuits are connected to a latch.
- Read/Write Latches are connected to data input register and data

Internal Organization of SDRAM



READ AND WRITE OPERATIONS

- For READ operation , the row address is applied first , and in response to the column address , the data present in the latches for the selected columns are transferred to the data output register.
  - Then the data is available on the data bus.

- For WRITE operation , the row address is applied first , and in response to the column address , the data present in the data bus is made available to the latches through data input register.
  - The data is then written to the particular cell.

BURST MODE TRANSFER

- As in fast page mode , after the column address is applied the data from successive column addresses are read out or written into.
- In same way burst operation of different lengths can be specified.
- This uses the same block transfer capability of fast page mode.

- Control signals , required are provided internally by column counter and clock signals.

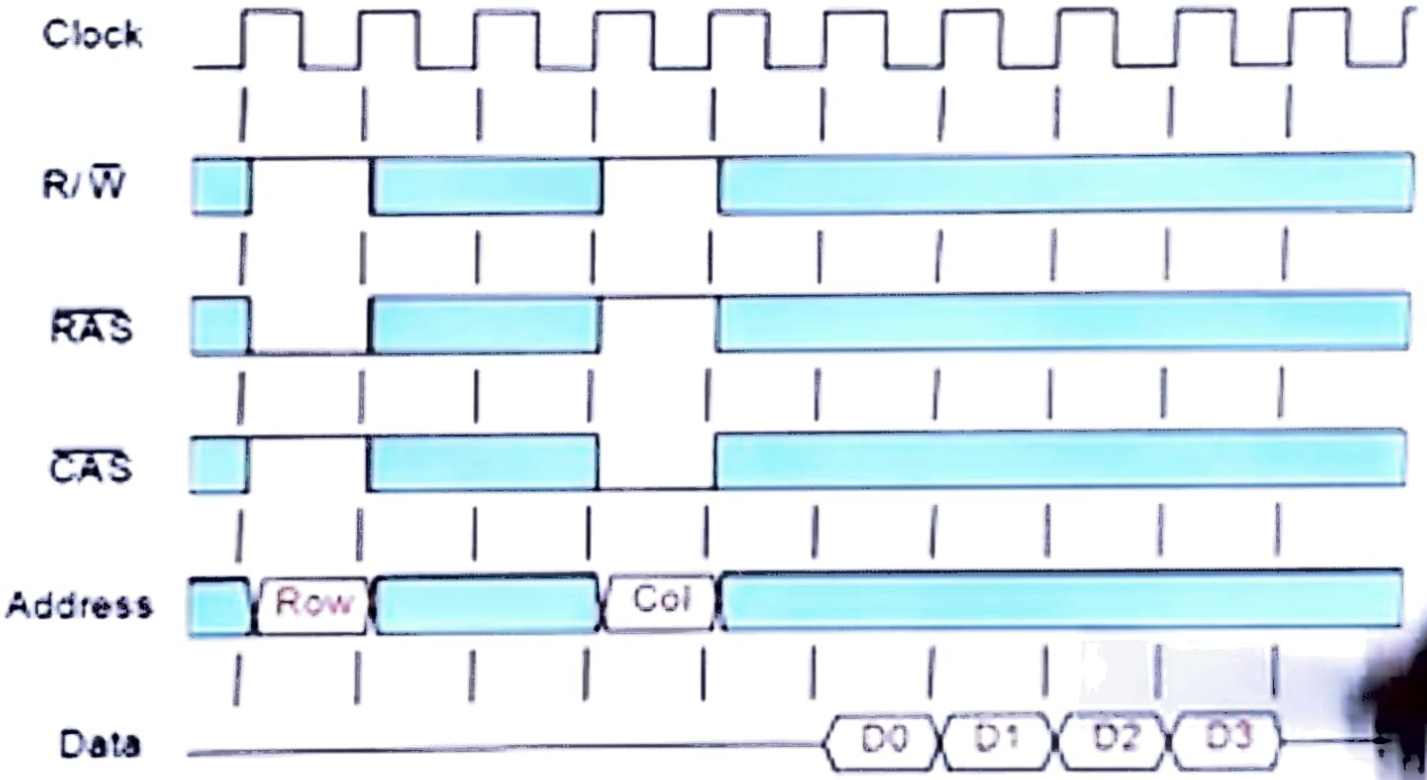
- CAS signal need not be generated externally. A new data is available in the data line after every clock cycle.

- All the operations are triggered during raising edge of the clock.

- It has built-in refresh circuitry, and refresh counter is part of it.

  During a refresh operation, the contents of the cells are refreshed

SDRAM TIMING DIAGRAM



- Row address is latched under control of RAS signal
- Memory takes about 2-3 cycles to activate selected row
- Column address is latched under control of CAS signal
- After delay of 1 cycle, first set of data bits placed on data lines

- SDRAM automatically increments column address to access the next 3 sets of bits in the selected row

LATENCY

- The speed and efficiency of data transfers among memory, processor, and disk have a large impact on the performance of a computer system.
- Memory latency – the amount of time it takes to transfer a word of data to or from the memory.
  - In block transfers, *latency* is used to denote the time it takes to transfer the first word of data
  - In previous diagram, access cycle begins with assertion of RAS and first word is transferred 5 cycles later.
- Hence latency is 5 clock cycles.

BANDWIDTH

- **Memory bandwidth** – the number of bits or bytes that can be transferred in one second. **It is used to** measure how much time is needed to transfer an entire block of data.
- **Depends on**:
  - Speed of memory access
  - Transfer capability of the links – speed of the bus
  - No of bits that can be accessed in parallel
- **Bandwidth is not determined solely by memory**. **It is the** product of the rate at which data are transferred (and accessed) and the width of the data bus.

STATIC VS DYNAMIC RAM

- Static RAM
- Bits stored as on/off switches via flip-flops
- No charges to leak
- No refreshing needed when powered
- More complex construction
- Larger per bit
- More expensive
- Does not need refresh circuits (volatile memory)
- Faster
- Cache

- Dynamic RAM
- Bits stored as charge in semiconductor capacitors
- capacitor charge leaks
- Need refreshing even when powered
- Simpler construction
- Smaller per bit
- Less expensive
- Need refresh circuits (every few milliseconds)
- Slower
- Main memory

8.2.5 Structure of Larger Memories

We have discussed the basic organization of memory circuits as they may be implemented on a single chip. Next, we examine how memory chips may be connected to form a much larger memory.

Static Memory Systems

Consider a memory consisting of 2M words of 32 bits each. Figure 8.10 shows how this memory can be implemented using 512K × 8 static memory chips. Each column in the figure implements one byte position in a word, with four chips providing 2M bytes. Four columns implement the required 2M × 32 memory. Each chip has a control input called

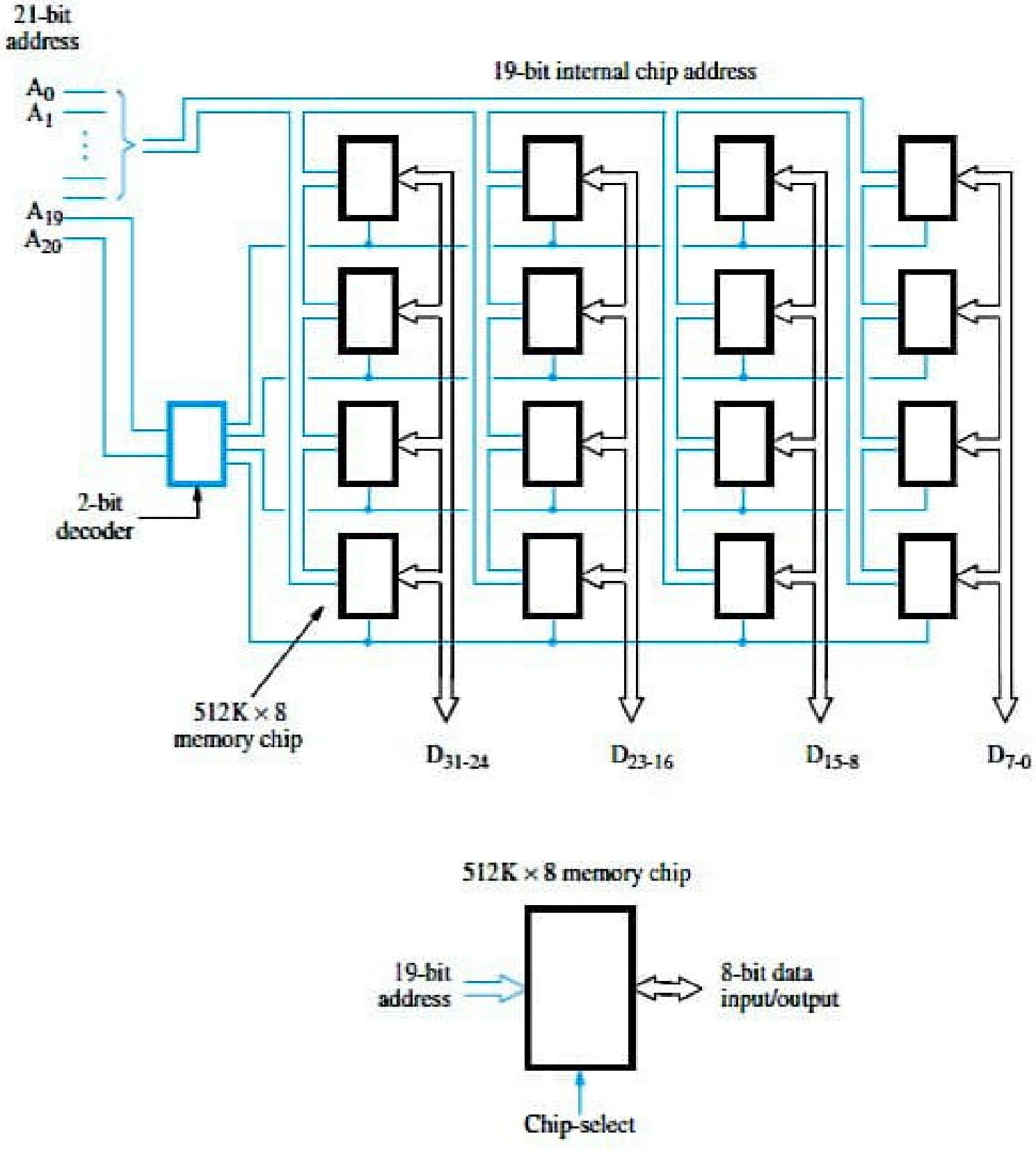


Figure 8.10 Organization of a 2M × 32 memory module using 512K × 8 static memory chips.

Chip-select. When this input is set to 1, it enables the chip to accept data from or to place data on its data lines. The data output for each chip is of the tri-state type described in Section 7.2.3. Only the selected chip places data on the data output line, while all other outputs are electrically disconnected from the data lines. Twenty-one address bits are needed to select a 32-bit word in this memory. The high-order two bits of the address are decoded to determine which of the four rows should be selected. The remaining 19 address bits are used to access specific byte locations inside each chip in the selected row. The R/$\overline{W}$ inputs of all chips are tied together to provide a common Read/$\overline{\text{Write}}$ control line (not shown in the figure).

Dynamic Memory Systems

Modern computers use very large memories. Even a small personal computer is likely to have at least 1G bytes of memory. Typical desktop computers may have 4G bytes or more of memory. A large memory leads to better performance, because more of the programs and data used in processing can be held in the memory, thus reducing the frequency of access to secondary storage.

Because of their high bit density and low cost, dynamic RAMs, mostly of the synchronous type, are widely used in the memory units of computers. They are slower than static RAMs, but they use less power and have considerably lower cost per bit. Available chips have capacities as high as 2G bits, and even larger chips are being developed. To reduce the number of memory chips needed in a given computer, a memory chip may be organized to read or write a number of bits in parallel, as in the case of Figure 8.7. Chips are manufactured in different organizations, to provide flexibility in designing memory systems. For example, a 1-Gbit chip may be organized as 256M $\times$ 4, or 128M $\times$ 8.

Packaging considerations have led to the development of assemblies known as memory modules. Each such module houses many memory chips, typically in the range 16 to 32, on a small board that plugs into a socket on the computer's motherboard. Memory modules are commonly called *SIMM*s (Single In-line Memory Modules) or *DIMM*s (Dual In-line Memory Modules), depending on the configuration of the pins. Modules of different sizes are designed to use the same socket. For example, 128M $\times$ 64, 256M $\times$ 64, and 512M $\times$ 64 bit DIMMs all use the same 240-pin socket. Thus, total memory capacity is easily expanded by replacing a smaller module with a larger one, using the same socket.

Memory controller

- Recall that in a dynamic memory chip, to reduce the number of pins, multiplexed addresses are used.
- Address is divided into two parts:
  - High-order address bits select a row in the array.
  - They are provided first, and latched using RAS signal.
  - Low-order address bits select a column in the row.
  - They are provided later, and latched using CAS signal.
- However, a processor issues all address bits at the same time.
- In order to achieve the multiplexing, memory controller circuit is inserted between the processor and memory.

Memory Controller

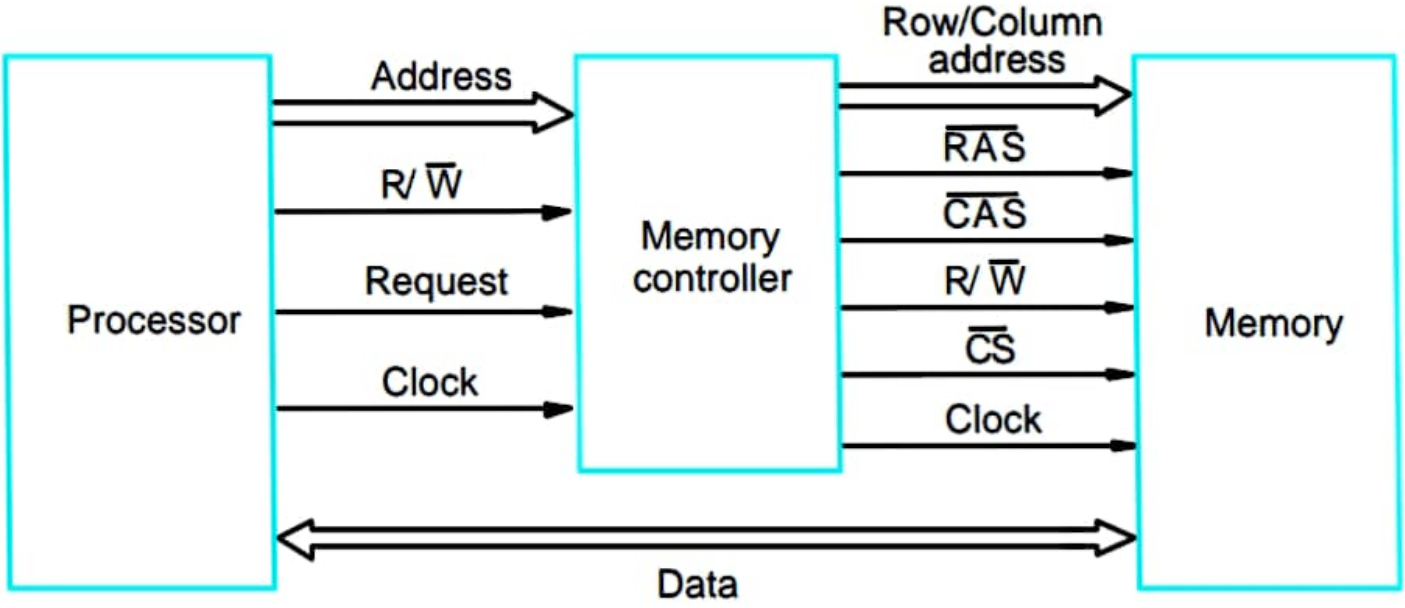


Figure 5.11. Use of a memory controller.

Refresh Overhead

A dynamic RAM cannot respond to read or write requests while an internal refresh operation is taking place. Such requests are delayed until the refresh cycle is completed. However, the time lost to accommodate refresh operations is very small. For example, consider an SDRAM in which each row needs to be refreshed once every 64 ms. Suppose that the minimum time between two row accesses is 50 ns and that refresh operations are arranged such that all rows of the chip are refreshed in 8K (8192) refresh cycles. Thus, it takes $8192 \times 0.050 = 0.41$ ms to refresh all rows. The refresh overhead is $0.41/64 = 0.0064$, which is less than 1 percent of the total time available for accessing the memory.

Choice of Technology

The choice of a RAM chip for a given application depends on several factors. Foremost among these are the cost, speed, power dissipation, and size of the chip.

Static RAMs are characterized by their very fast operation. However, their cost and bit density are adversely affected by the complexity of the circuit that realizes the basic cell. They are used mostly where a small but very fast memory is needed. Dynamic RAMs, on the other hand, have high bit densities and a low cost per bit. Synchronous DRAMs are the predominant choice for implementing the main memory.

8.3.1 ROM

A memory is called a read-only memory, or ROM, when information can be written into it only once at the time of manufacture. Figure 8.11 shows a possible configuration for a ROM cell. A logic value 0 is stored in the cell if the transistor is connected to ground at point P; otherwise, a 1 is stored. The bit line is connected through a resistor to the power supply. To read the state of the cell, the word line is activated to close the transistor switch. As a result, the voltage on the bit line drops to near zero if there is a connection between the transistor and ground. If there is no connection to ground, the bit line remains at the high voltage level, indicating a 1. A sense circuit at the end of the bit line generates the proper output value. The state of the connection to ground in each cell is determined when the chip is manufactured, using a mask with a pattern that represents the information to be stored.

8.3.2 PROM

Some ROM designs allow the data to be loaded by the user, thus providing a *programmable ROM* (PROM). Programmability is achieved by inserting a fuse at point P in Figure 8.11. Before it is programmed, the memory contains all 0s. The user can insert 1s at the required locations by burning out the fuses at these locations using high-current pulses. Of course, this process is irreversible.

PROMs provide flexibility and convenience not available with ROMs. The cost of preparing the masks needed for storing a particular information pattern makes ROMs cost-effective only in large volumes. The alternative technology of PROMs provides a more convenient and considerably less expensive approach, because memory chips can be programmed directly by the user.

8.3.3 EPROM

Another type of ROM chip provides an even higher level of convenience. It allows the stored data to be erased and new data to be written into it. Such an *erasable*, reprogrammable ROM is usually called an *EPROM*. It provides considerable flexibility during the development phase of digital systems. Since EPROMs are capable of retaining stored information for a long time, they can be used in place of ROMs or PROMs while software is being developed. In this way, memory changes and updates can be easily made.

An EPROM cell has a structure similar to the ROM cell in Figure 8.11. However, the connection to ground at point P is made through a special transistor. The transistor is normally turned off, creating an open switch. It can be turned on by injecting charge into it that becomes trapped inside. Thus, an EPROM cell can be used to construct a memory in the same way as the previously discussed ROM cell. Erasure requires dissipating the charge trapped in the transistors that form the memory cells. This can be done by exposing the chip to ultraviolet light, which erases the entire contents of the chip. To make this possible, EPROM chips are mounted in packages that have transparent windows.

8.3.4 EEPROM

An EPROM must be physically removed from the circuit for reprogramming. Also, the stored information cannot be erased selectively. The entire contents of the chip are erased when exposed to ultraviolet light. Another type of erasable PROM can be programmed, erased, and reprogrammed electrically. Such a chip is called an *electrically erasable* PROM, or EEPROM. It does not have to be removed for erasure. Moreover, it is possible to erase the cell contents selectively. One disadvantage of EEPROMs is that different voltages are needed for erasing, writing, and reading the stored data, which increases circuit complexity. However, this disadvantage is outweighed by the many advantages of EEPROMs. They have replaced EPROMs in practice.

- Flash memory:
  - Has similar approach to EEPROM.
  - Read the contents of a single cell, but write the contents of an entire block of cells.
  - Flash devices have greater density. Higher capacity and low storage cost per bit.
  - Power consumption of flash memory is very low, making it attractive for use in equipment that is battery-driven.
  - Single flash chips are not sufficiently large, so larger memory modules are implemented using flash cards and flash drives.